



Centrus™ Spatial Transforms User's Guide

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Preface

Who Should Read this Manual

This manual is intended for users of Design Studio and Information Studio who create Plans to do Point-in-Polygon (PIP) and Closest Site analyses.

To use the Sagent Information Studio, you should be familiar with your Spatial data, know how to create Plans and work with Transforms, and have permission to work in the Data Flow Editor. For more information on working with Plans and Transforms, refer to *Sagent Information Studio, Getting Started* and the *Sagent Design Studio User's Guide*.

Conventions

The following conventions are used in this document:

- Keys on the keyboard are referred to by name, such as Escape and Enter.
- Commands you type or items you select with the mouse are printed in **bold**, such as clicking **Update** or typing **yes**.
- System messages and text on your screen are printed in a different font, such as the *Filter Type* field.

The cursor changes depending on actions that you are performing. Some examples are:

	Drag cursor, appears when you can drag and drop an item to a new location.
	No drop cursor, appears when the item you are dragging cannot be dropped.
	Handle cursor, appears when you can move a divider on a screen.

How to Proceed

This manual describes how to use the Point-in-Polygon and Closest Site Transforms (collectively referred to as the Spatial Transforms).

Use the following summary to determine which chapters you need to read:

Chapter 1	explains how the Spatial Transforms work, the analyses these Transforms make, and some various concepts needed to use the Transforms.
Chapter 2	provides a comprehensive reference for the dialogs and options available in the Spatial Transforms.
Chapter 3	provides examples of how to use the Spatial Transforms.
Appendix A	explains how to use the GSBimp utility.
Appendix B	gives an overview of the ESRI BNA file format.
Appendix C	summarizes the subset of MapInfo's Data Interchange Format known as the MIF/MID format supported by the Point-in-Polygon Transform.
Glossary	definitions for Spatial analysis terms.

Chapter 1

Introduction

This chapter explains how the Spatial Transforms work, the analyses these Transforms make, and some various concepts needed to use the Transforms.

In this chapter:

- Spatial Analysis
- Point-in-Polygon Analysis
- Closest Site Analysis
- Coordinates
- Spatial File Types

Spatial Analysis

The value of a geocoded address does not come from the latitude and longitude data itself, but from the business information that results from comparing a spatial point to another spatial entity. For example, the spatial question “Does this point fall in this polygon?” answers the business question “Does this customer live in an area we service?” The resulting answer allows for informed decision making. Spatial analysis describes the relationship in space between two entities. For the purpose of these Transforms, a spatial relationship is typically between objects that could be placed on a map (such as stores, houses, shorelines, and cities.)

A Point-in-Polygon analysis lets you accurately determine within which areas a given point falls. Retail, insurance, and telecommunication industries use Point-in-Polygon analyses to identify key markets, insurance risk areas, and telephone service areas.

A Closest Site analysis provides the closest distance between a point and a set of other points. The healthcare, banking, and insurance industries use this type of analysis to match up doctors and patients, find the nearest bank branch to a customer, and calculate the distance between a home and a fire station.

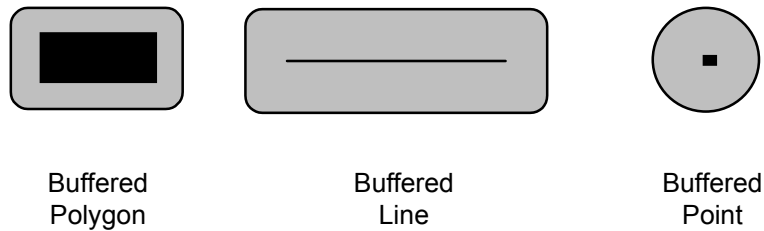
Point-in-Polygon Analysis

A PIP analysis determines in which polygon (or polygons) a point falls. Examples of PIP analyses are:

- Assigning sales territory for new customer records.
- Calculating which store trade areas contain the most overlap, by measuring the amount of customer overlap rather than area of overlap.

The PIP Transform “buffers” a line, point, or polygon. The area enclosed by the buffer is treated like a polygon, which can then be searched to determine if a point falls inside. Figure 1 shows some examples of buffered objects.

Figure 1: Buffer types



Collectively, all of these items are referred to as “objects”.

Buffering allows you to solve problems such as determining if a point (or points) is within 500 feet of a shoreline, within two miles of a store, or within 1.3 miles of a sales territory.

Closest Site Analysis

Closest Site analyses are generally performed on two sets of geocoded addresses. Commonly, one set of addresses represents store locations, while another set of addresses represents current or potential customers.

In other words, a Closest Site analysis is a distance calculation between two sets of points. Examples of Closest Site analyses are:

- Determining which store is closest to each customer.
- Listing the five closest doctors to a potential client.
- Listing up to eight store locations within 15 miles of a proposed site.

Coordinates

The Spatial Transforms require coordinates to an accuracy of a millionth of a degree. The coordinates are entered with a multiplier so that the product of the coordinate and multiplier is an integer.

For example, the ACC Transform outputs a pair of latitude/longitude values as -105.160311, 40.060177 to be used by the Spatial Transform. Multiplying these values with a multiplier of 1,000,000 produces the coordinate pair -105160311, 40060177. If the final values are not in this format, the output of the Spatial Transforms will be unreliable. The Spatial Transforms always assume that longitude is the horizontal direction and latitude is the vertical direction.

In the Spatial Transforms, positive longitude values represent locations east of the prime meridian, and positive latitude values represent locations north of the equator. You can use the sign of the multiplier to change initial coordinates to match this scheme. For example, you could enter a positive longitude that represents the Southern hemisphere, but use a negative multiplier to create a final negative longitude value.

The multiplier is used for internal calculations in the Spatial Transforms, and does not change the original coordinate values in the input Data Flow.

Note: Latitude/longitude values may be passed in to the Spatial Transforms as either string or numeric data types. If the values are passed in as strings, Transform performance is impaired.

Spatial File Types

The PIP Transform requires Spatial files in a proprietary GSB format.

GSB files can be created from ESRI Atlas GIS BNA files, MapInfo MIF/MID files, or ESRI SHP files. The formats for the BNA and MIF/MID file types are published and are provided, as supported by the Spatial Transforms, in Appendix B, “ESRI BNA Format” and Appendix C, “MapInfo MIF/MID Format”.

The Spatial Transforms package includes a utility for creating GSB files from these other formats. See Appendix A, “GSBimp Utility” on page 31 for instructions on using the utility.

When importing a BNA, MIF/MID, or SHP file, all valid objects are stored in the object file. Valid objects are points, lines, or polygons that have been properly defined. Unsupported objects, such as circles or malformed objects, are not stored.

A MIF/MID file is actually two files, one with the extension .mif, and the other with the extension .mid. The two MIF and MID files must have the same base name, and exist in the same directory, in order for the GSBimp utility to convert them to a single GSB file.

ESRI shape files include DBF, SHP, and SHX files. SHP files include points and geometric information. DBF files contain attributes and names of polygons. SHX files contain indexes for faster performance. The DBF, SHP, and SHX files must have the same base name, and exist in the same directory, in order for the GSBimp utility to convert them to a single GSB file.

Chapter 2

Using the Spatial Transforms

This chapter provides a comprehensive reference for the dialogs and options available in the Spatial Transforms.

In this chapter:

- Using the Point-in-Polygon Transform
- Using the Closest Site Transform

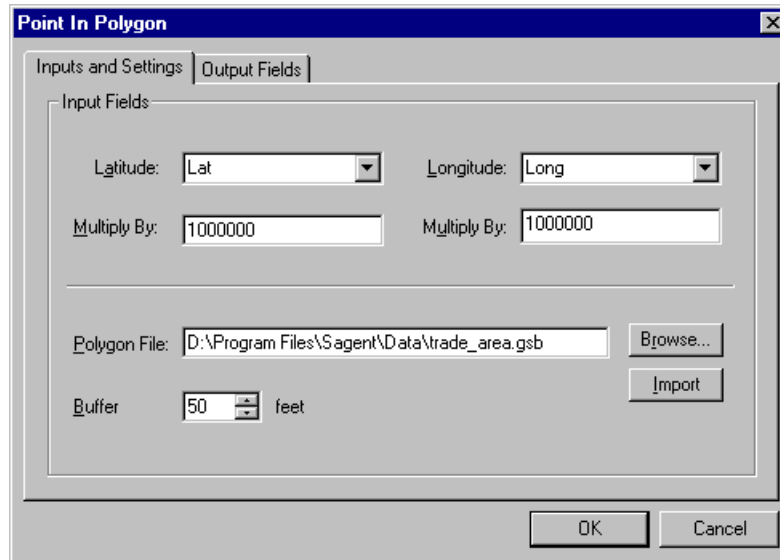
Using the Point-in-Polygon Transform

To perform PIP analyses, you must first create an object file. Object files can be created from ESRI Atlas GIS BNA files, MapInfo MIF/MID files, or ESRI SHP files.

To perform PIP functions, the data files need to have latitude/longitude coordinates assigned to each point. These assignments can be made through the Address Cleanser & Coder Transform.

Inputs and Settings

Use this tab to assign the latitude and longitude input fields, specify the latitude and longitude multipliers, choose the polygon input file, and set the polygon buffer width.



Latitude and Longitude

These two dropdown lists are used to select the fields from the input data file that contain the latitude and longitude values.

If the input data file can not be read by the Data Flow, you must type the name of the field that contains the latitude or longitude value, instead of selecting it from a list.

The **Multiply By** values are used to adjust the initial latitude and longitude values so that they are usable by the Transform.

Polygon File

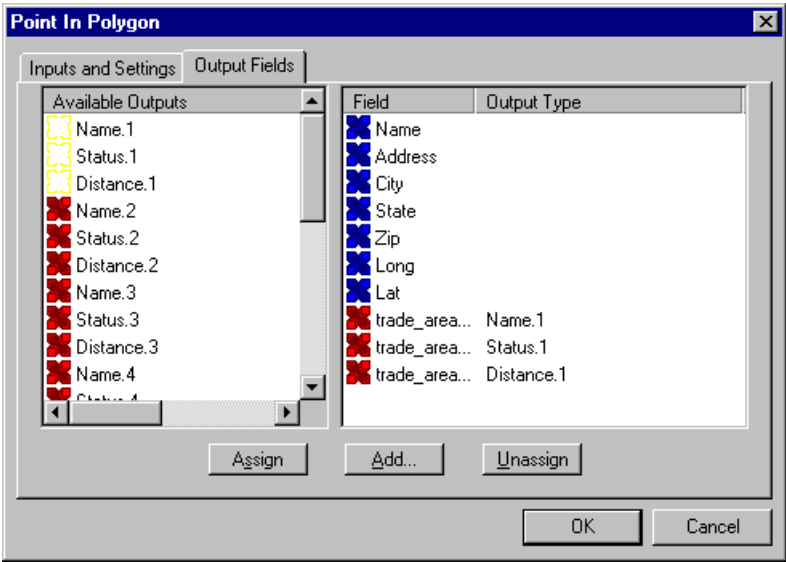
Use this text box to choose the polygon data file (.gsb file). The path must be relative to the Data Flow Server machine. If you have a polygon file that is not in the GSB format, you can click **Import**, which will launch the GSBimp utility. See Appendix A, "GSBimp Utility" for details on how to use this utility.

Buffer

The **Buffer** text box is used to set the buffer width (in feet) around the polygons. For example, if you set the Buffer to 10 feet, every polygon in the file will have a buffer of 10 feet on each side of the line (for a total buffer width of 20 feet).

Output Fields

Use this dialog to assign output data to individual output fields. Each object within a GSB file is associated with three output data elements:



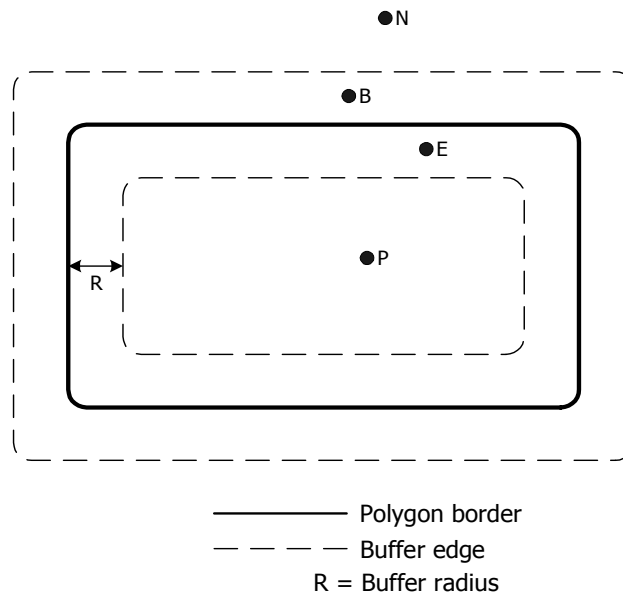
- **Name** is the name of the feature as it is defined in the layer.
- **Status** defines where a point lies in relation to the polygon. Status codes are:

P = In polygon

E = Edge; in buffer, inside polygon

B = In buffer, outside polygon

N = Not found



- **Distance** is the distance (in feet) from the point to the edge of the polygon.

The PIP Transform returns information for up to eight polygons. The objects in the GSB file must be built with buffering enabled in order to return the Edge and In buffer status values, and the distance value. If the GSB file does not have buffering enabled, the distance value will be blank.

Assigning Output Data to Data Flow Fields

To assign an available output data element to an existing Data Flow field:

1. Click an element name in the **Available Outputs** column.
2. Click a field name from the **Field** column.
3. Click **Assign**.

The selected output type appears in the Output Type column next to the field name to which it was assigned.

Note: You can also assign available output data elements to existing Data Flow fields by dragging the available output element onto the field name.

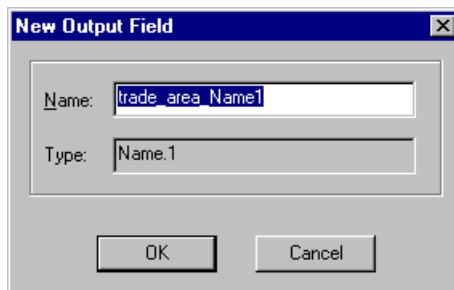
If the input Data Flow can not be read, and you want to assign data to a field that already exists (but is not displayed), follow the steps for adding a new field and type the name of the existing field. When the Data Flow is passed to subsequent steps, this field will be populated with the assigned data.

Adding New Fields

To create a new output field:

1. Click an Available Output element.
2. Click **Add**.

The New Output Field dialog displays.



You can also open the New Output Field dialog by double-clicking an available output type.

3. Type a field name and click **OK**, or accept the default field name by clicking **OK**.

The new field name appears in the Field column with the output type.

Unassign Fields

If the currently selected field is an input field, clicking **Unassign** removes its output assignment. If the currently selected field is a new field, clicking **Unassign** deletes it.

Note: You can also unassign an output field by double-clicking it.

Using the Closest Site Transform

To do a Closest Site analysis requires two Data Flows with geocoded points. One Data Flow contains the points to be tested, and the other Data Flow contains points to be tested against. When creating your plan, the top Data Flow must contain the input location data, and the bottom flow must contain the site data.

The Closest Site routines within the Closest Site Transform allow you to find points that are within a specified search distance. A maximum of eight points can be returned, in order of closest to farthest.

Input Locations

Use the Input Locations tab to assign the latitude and longitude fields for the input locations Data Flow (the top Data Flow into the Transform), specify the latitude and longitude multipliers, and set the search radius.

The columns from the Data Flow specified here will be made available under the Output Assignments tab and will be passed on to other steps in the plan.

The screenshot shows the 'Closest Site' dialog box with the 'Input Locations' tab selected. The dialog has three tabs: 'Input Locations', 'Site Locations', and 'Output Assignments'. The 'Input Location' section contains two columns of settings. The first column has 'Latitude:' with a dropdown menu showing 'Lat' and 'Multiply By:' with a text box containing '1000000'. The second column has 'Longitude:' with a dropdown menu showing 'Long' and 'Multiply By:' with a text box containing '1000000'. Below these is a 'Max Distance (Miles) To Site' section with a text box containing '5'. At the bottom right are 'OK' and 'Cancel' buttons.

Field	Value
Latitude	Lat
Longitude	Long
Multiply By (Latitude)	1000000
Multiply By (Longitude)	1000000
Max Distance (Miles) To Site	5

Input Location

The **Latitude** and **Longitude** dropdown lists are used to select the fields from the input locations Data Flow that contain the latitude and longitude values.

If the input data file can not be read by the Data Flow, you must type the name of the field instead of selecting it from a list.

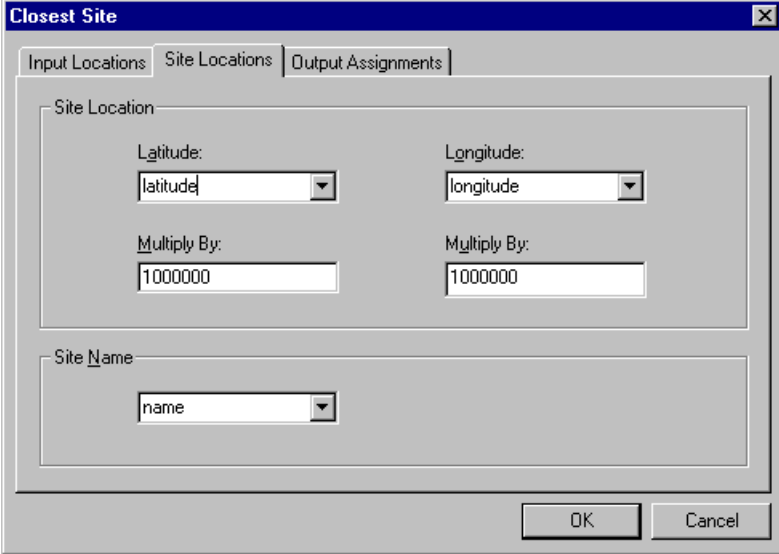
The **Multiply By** values are used to adjust the initial latitude and longitude values so that they are usable by the Transform.

Max Distance (Miles) To Site

Use this text box to specify the search radius around the input locations.

Site Locations

Use the Site Locations tab to assign the latitude, longitude, and site name fields for the site locations Data Flow (the bottom Data Flow into the Transform), and to specify the latitude and longitude multipliers.



The screenshot shows the 'Closest Site' dialog box with the 'Site Locations' tab selected. The dialog has three tabs: 'Input Locations', 'Site Locations', and 'Output Assignments'. The 'Site Location' section contains two columns of settings. The first column has 'Latitude:' with a dropdown menu showing 'latitude' and 'Multiply By:' with a text box containing '1000000'. The second column has 'Longitude:' with a dropdown menu showing 'longitude' and 'Multiply By:' with a text box containing '1000000'. Below this, the 'Site Name' section has a dropdown menu showing 'name'. At the bottom right are 'OK' and 'Cancel' buttons.

Field	Value
Latitude	latitude
Longitude	longitude
Multiply By (Latitude)	1000000
Multiply By (Longitude)	1000000
Site Name	name

Site Location

The **Latitude** and **Longitude** dropdown lists are used to select the fields from the site locations Data Flow that contain the latitude and longitude values.

If the input data file can not be read by the Data Flow, you must type the name of the field instead of selecting it from a list.

The **Multiply By** values are used to adjust the initial latitude and longitude values so that they are usable by the Transform.

Site Name

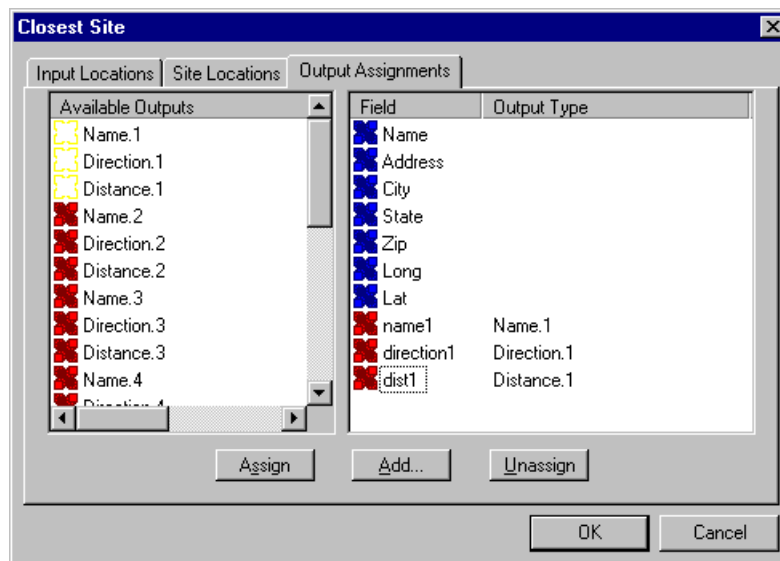
Select the field from the dropdown list that contains the site name values for the site locations Data Flow. These values will be returned as Name output in the Output Assignments tab.

If the input data file can not be read by the Data Flow, you must type the name of the field instead of selecting it from a list.

Output Assignments

Use the Output Assignments tab to assign output data to individual output fields. The fields displayed in the Field column are associated with the Data Flow specified in the Input Locations tab.

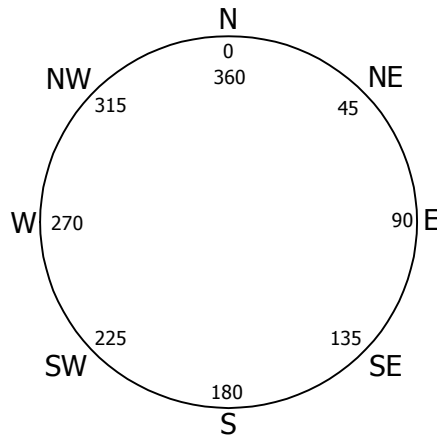
Each feature within a Closest Site layer is associated with three output data elements:



- **Name** is the name of the feature as it is defined in the layer.
- **Direction** is the direction from the address to the closest site, measured in degrees, with north at 0/360 degrees.

Figure 2 shows how degree output from the Closest Site Transform relates to compass North, South, East, and West.

Figure 2: Direction output from the Closest Site Transform



- **Distance** is the distance (in miles) from the address to the closest site in the layer.

The sites are returned in order of nearest to farthest. Up to eight sites are returned for each input location.

Assigning Output Data to Data Flow Fields

To assign an available output data element to an existing Data Flow field:

1. Click an element name in the **Available Outputs** column.
2. Click a field name from the **Field** column.
3. Click **Assign**.

The selected output type appears in the Output Type column next to the field name to which it was assigned.

Note: You can also assign available output data elements to existing Data Flow fields by dragging the available output element onto the field name.

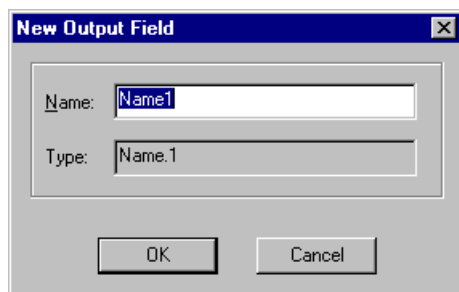
If the input Data Flow can not be read, and you want to assign data to a field that already exists (but is not displayed), follow the steps for adding a new field and type the name of the existing field. When the Data Flow is passed to subsequent steps, this field will be populated with the assigned data.

Adding New Fields

To create a new output field:

1. Click an Available Output element.
2. Click **Add**.

The New Output Field dialog displays.



You can also open the New Output Field dialog by double-clicking an available output type.

3. Type a field name and click **OK**, or accept the default field name by clicking **OK**.

The new field name appears in the Field column with the output type.

Unassign Fields

If the currently selected field is an input field, clicking **Unassign** removes its output assignment. If the currently selected field is a new field, clicking **Unassign** deletes it.

Note: You can also unassign an output field by double-clicking it.

Chapter 3

Tutorials

This chapter provides examples of how to use the Spatial Transforms.

In this chapter:

- Overview
- Point-in-Polygon Transform Tutorial
- Closest Site Transform Tutorial

Overview

Both of these tutorials use the premise that you are a coffee shop mogul, and are considering opening new stores in Boulder, Colorado. You have the following files:

- A polygon file that contains the trade areas for the proposed stores (`trade_area.gsb`).
- A site file that contains the latitude/longitude coordinates for the proposed store sites (`stores.csv`).
- A prospect file that contains the latitude/longitude coordinates for the prospect customers (`sample_cust.csv`).

In the PIP tutorial, you determine which customers fall into which proposed trade area. In the Closest Site tutorial, you finish the analysis by calculating the shortest distance from each customer to one of your new coffee shops.

Point-in-Polygon Transform Tutorial

Point-in-Polygon analysis determines within which object, or objects, a point falls. The Point-in-Polygon Transform can perform analyses on polygons, as well as on buffered lines, buffered points, and buffered polygons. In this example, use a Point-in-Polygon analysis to determine which customers fall within certain store trade areas.

Source Data

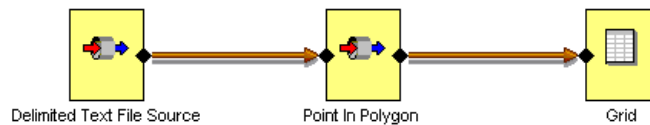
The address information used in this example is a list hypothetical customers. This data set resides in the file `sample_cust.csv`.

	A	B	C	D	E	F	G
1	Name	Address	City	State	Zip	Long	Lat
2	Abbott B G	8100 KINCROSS DR	BOULDER	CO	80301	-105.160311	40.060177
3	Adams Metha	4833 WHITE ROCK CIR	BOULDER	CO	80301	-105.211278	40.061439
4	P J Ahrens	4641 PORTSIDE WAY	BOULDER	CO	80301	-105.200718	40.059194
5	Ahuna D L	6200 HABITAT DR	BOULDER	CO	80301	-105.208946	40.060746
6	Albright Lloyd M	3889 PASEO DEL PRADO ST	BOULDER	CO	80301	-105.256221	40.042906

Building a Plan

1. Open the Data Flow Editor.
2. Open the Tool Bin.
3. Drag these Transforms into the Data Flow Editor in the following order:
 - Delimited Text File Source
 - Point In Polygon
 - Grid

Your Data Flow appears as follows:



Accessing Source Data

1. Double-click the Delimited Text File Source Transform.
2. Click **Browse** in the Sample Data section of the Delimited Text File Source dialog.
3. Browse to the location of your source data file, select `sample_cust.csv`, and click **Open**.

The column names in your source file appear in the list field below **Browse**.

4. Populate the Field List by clicking **Column Names** and **Data Types** in the Auto-Populate Field List section.

The columns display in the Field List.

5. Copy the full path and file name of the source file from the File Name field.
6. Paste the full path and file name of the source file into the File Name on Server field. If necessary, modify the path to be local to the Data Flow Server Machine.

Note: Select the **Column Names in File** option if you want the column names to appear in the column headers in the Grid.

The settings in the Delimited Text File Source dialog appear as follows:

Delimited Text File Source

File Attributes

File Name On Server: tion\Transform Spatial\sample_cust.csv

Character Set: ANSI

Separator Character: ,

Quote Character: "

Comment Character: #

☒ Column Names In File

☒ Extended Character Set

Field List

Add... Remove

Col No	Column Name	Output Type	Length	Scale
1	Name	String	23	
2	Address	String	24	
3	City	String	8	
4	State	String	3	
5	Zip	Integer		
6	Long	Double		
7	Lat	Double		

Auto-Populate Field List

Column Names

Data Types

Sample Records To Scan: (0=All) 100

☐ Populate at runtime

Sample Data

File Name: tion\Transform Spatial\sample_cust.csv Browse....

	Name	Address	City	State	Zip	Long	Lat
Row1	Abb...	8100 ...	BO...	CO	80301	-105...	40...
Row2	Ada...	4833 ...	BO...	CO	80301	-105...	40...
Row3	P J ...	4641 ...	BO...	CO	80301	-105...	40...
Row4	Ahu...	6200 ...	BO...	CO	80301	-105...	40...
Row5	Albri...	3889 ...	BO...	CO	80301	-105...	40...
Row6	Mr I...	2995 ...	BO...	CO	80301	-105...	40...

OK Cancel

7. Click **OK**.

The Column Names of your customer information display in the Grid.

Point In Polygon Information

1. Double-click the Point-in-Polygon Transform.
2. Click the Inputs and Settings dialog.
3. Select the fields that contain the latitude and longitude values from the **Latitude** and **Longitude** drop-down lists.
4. Leave the latitude and longitude multipliers at their default values.
5. Click **Browse**, and find the location of `trade_area.gsb`.
6. Select the polygon file and click **Open**.
7. Leave **Buffer** at the default value.
8. Click the Output Fields tab.
 The column names from the Data Flow file display in the Field column. The blue icon next to the column name indicates that the column is unassigned.
9. Click **Name.1** from the Available Outputs list, and click **Add**.
10. Accept the default name by clicking **OK**. Notice that the final output field name begins with the Polygon File name.
11. You have just added the **Name.1** field as the output field **trade_area_Name1**. Now add the **Status.1** and **Distance.1** fields as output fields.
12. Click **OK**.
 The new columns display in the Grid.
13. Click **Update**.
14. Save the Plan as `PIPEExample1`.

Viewing the Results

The following Grid shows the added output fields of the Point-in-Polygon analysis.

trade_area_Name1	trade_area_Status1	trade_area_Distance1
Trade_Area 130	P	593
Trade_Area 130	P	2386
Trade_Area 130	P	5108
Trade_Area 130	P	3058
Trade_Area 120	P	1761
Trade_Area 110	P	1472
Trade_Area 140	P	2786
Trade_Area 110	P	1919

The trade_area_Name1 field contains the polygon identifier. The trade_area_Status1 field identifies where the point is in relation to the object (in this case, "P" indicates the point is inside the polygon). The trade_area_Distance1 field contains the minimum distance between the point and the trade area border.

Closest Site Transform Tutorial

Closest Site analysis compares the points in your input location file to those in the site file. The Closest Site Transform identifies the closest sites for each point in the input file. For each site found, the site identifier, straight-line distance, and direction (in degrees) to the site can be returned. In this example, use Closest Site analysis to find store locations near your customers.

Source Data

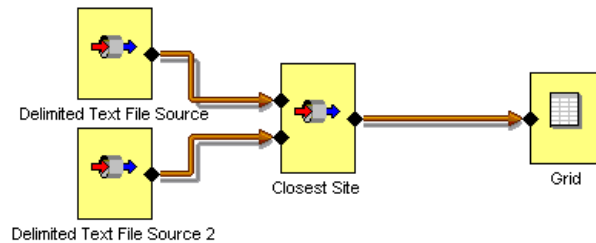
The address information used in this example is a list hypothetical customers. This data set resides in the file `sample_cust.csv`.

	A	B	C	D	E	F	G
1	Name	Address	City	State	Zip	Long	Lat
2	Abbott B G	8100 KINCROSS DR	BOULDER	CO	80301	-105.160311	40.060177
3	Adams Metha	4833 WHITE ROCK CIR	BOULDER	CO	80301	-105.211278	40.061439
4	P J Ahrens	4641 PORTSIDE WAY	BOULDER	CO	80301	-105.200718	40.059194
5	Ahuna D L	6200 HABITAT DR	BOULDER	CO	80301	-105.208946	40.060746
6	Albright Lloyd M	3889 PASEO DEL PRADO ST	BOULDER	CO	80301	-105.256221	40.042906

Building a Plan

1. Open the Data Flow Editor.
2. Open the Tool Bin.
3. Drag these Transforms into the Data Flow Editor in the following order:
 - Delimited Text File Source
 - Delimited Text File Source
 - Closest Site
 - Grid

Your Data Flow should appears as follows:



Accessing Source Data

1. Rename the top Delimited Text File Source by right-clicking it, choosing **Rename Step**, and typing `Customer Data Flow`.
2. Rename the bottom Delimited Text File Source by right-clicking it, choosing **Rename Step**, and typing `Store Data Flow`.
3. Double-click `Customer Data Flow`.
4. Click **Browse** in the Sample Data section of the Delimited Text File Source dialog.

5. Browse to the location of your source data file, select `sample_cust.csv`, and click **Open**.

The column names in your source file appear in the list field below the Browse button.

6. Populate the Field List by clicking **Column Names** and **Data Types** in the Auto-Populate Field List section.

The columns display in the Field List.

7. Copy the full path and file name of the source file from the File Name field.

8. Paste the full path and file name of the source file into the File Name on Server field. If necessary, modify the path to be local to the Data Flow Server Machine.

Note: Select the **Column Names in File** option if you want the column names to appear in the column headers in the Grid.

The settings in the Delimited Text File Source dialog appear as follows:

Delimited Text File Source

File Attributes

File Name On Server: Character Set: Separator Character: ☒ Column Names In File

Quote Character: ☒ Extended Character Set

Comment Character:

Field List

Add ... Remove

Col No	Column Name	Output Type	Length	Scale
1	Name	String	23	
2	Address	String	24	
3	City	String	8	
4	State	String	3	
5	Zip	Integer		
6	Long	Double		
7	Lat	Double		

Auto-Populate Field List

Sample Records To Scan: (0=All)

☐ Populate at runtime

Sample Data

File Name:

	Name	Address	City	State	Zip	Long	Lat
Row1	Abb...	8100 ...	BO...	CO	80301	-105...	40...
Row2	Ada...	4833 ...	BO...	CO	80301	-105...	40...
Row3	P J ...	4641 ...	BO...	CO	80301	-105...	40...
Row4	Ahu...	6200 ...	BO...	CO	80301	-105...	40...
Row5	Albri...	3889 ...	BO...	CO	80301	-105...	40...
Row6	Mr I...	2995 ...	BO...	CO	80301	-105...	40...

9. Click **OK**.

The Column Names of your customer information display in the Grid.

10. Double-click the Store Data Flow Source.

11. Click **Browse** in the Sample Data section of the Delimited Text File Source dialog.
12. Browse to the location of your source data file, select `stores.csv`, and click **Open**.
13. The column names in your source file appear in the list field below **Browse**.
14. Populate the Field List by clicking **Column Names** and **Data Types** in the Auto-Populate Field List section.
15. The columns display in the Field List.
16. Copy the full path and file name of the source file from the File Name field.
17. Click **OK**.

Closest Site Information

1. Double-click the Closest Site Transform.
2. Click the Inputs Locations tab. This area refers to the Customer Data Flow Source.
3. Select the fields that contain the latitude and longitude values from the **Latitude** and **Longitude** drop-down lists.
4. Leave the latitude and longitude multipliers at their default values.
5. Leave the **Max Distance** at five miles (the default).
6. Click the Site Locations tab. This area refers to the Store Data Flow Source.
7. Select the fields that contain the latitude and longitude values from the **Latitude** and **Longitude** drop-down lists.
8. Leave the latitude and longitude multipliers at their default values.
9. Select the Site Name field from the drop-down list. This field will hold the object identifier.

10. Click the Output Assignments tab.

The column names from the Customer Data Flow Source are displayed in the Field column. The blue icon next to the column name indicates that the column is unassigned.

11. Click on **Name.1** from the Available Outputs list, and click **Add**.

12. Accept the default name by clicking **OK**.

You have just added the **Name.1** field as an output field.

13. Add the **Direction.1** and **Distance.1** fields as output fields.

14. Click **OK**.

The new columns display in the Grid.

15. Click **Update**.

16. Save the Plan as `ClosestSiteExample1`.

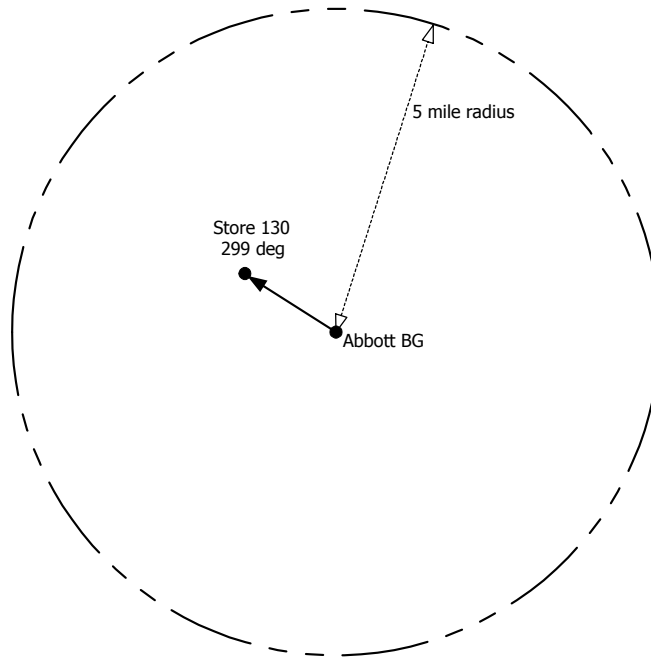
Viewing the Results

The following Grid shows the added output fields of the Closest Site analysis.

name1	direction1	dist1
Store 130	299	1.541
Store 130	64	1.489
Store 130	44	1.123
Store 130	60	1.402
Store 110	149	1.037
Store 110	117	0.429
Store 140	80	0.413
Store 110	188	0.390

The name1 field contains the polygon identifier. The direction1 field identifies the direction in degrees from the point to the object (see Figure 3 on page 30). The dist1 field contains the distance between the two points.

Figure 3: Representation of first customer and nearest store



Appendix A

GSBimp Utility

This appendix explains how to use the GSBimp utility.

In this appendix:

- Using the GSBimp Utility

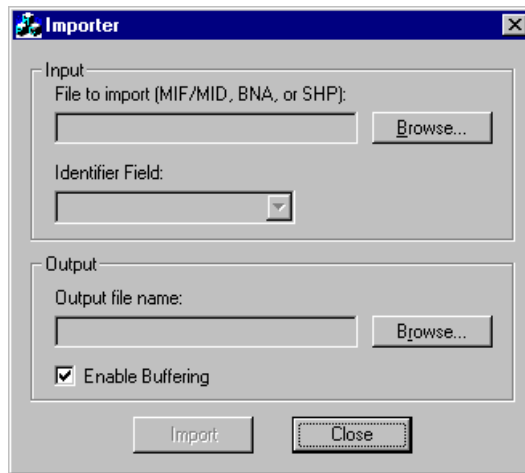
In order to use the PIP Transform, you must have a Spatial data file in the GSB format. The GSBimp utility allows you to convert other polygon file formats (BNA, MIF/MID, and SHP) to the GSB format.

Using the GSBimp Utility

1. Double click the Point-in-Polygon Transform.
2. From the Inputs and Settings tab, click **Import**.

Note: You can also run the GSBimp utility directly by executing the GSBImp .exe file included with the Spatial Transforms installation.

The Importer dialog appears.



3. Click **Browse** to select the .mif, .bna, or .shp file to import.
4. Select the field that contains the object identifier from the **Identifier Field** dropdown list.
5. To create the output file, click **Browse**. Browse to the appropriate directory and type the file name in the **File name** text box.
6. If you select **Enable Buffering**, the PIP Transform can return distances from the point to a polygon edge, and can determine if the point lies within the buffer or border of a polygon. If you do not select **Enable Buffering**, this information will not be available.
7. Click **Import** to finish.

Appendix B

ESRI BNA Format

This appendix gives an overview of the ESRI BNA file format.

In this appendix:

- Introduction
- Region Definitions
- Line Definitions
- Point Definitions

Introduction

The Point-in-Polygon Transform supports ESRI's Atlas GIS BNA format (also referred to as the Atlas ASCII format.) This section summarizes the format as supported by the PIP Transform. For a more complete description of the format, contact ESRI.

The PIP Transform supports Region, Line, and Point type objects from a BNA file. Circles are ignored and not imported.

A BNA object must have at least one identifier, and can have up to three identifiers. Each object must have the same number of identifiers, even if they are blank strings. In BNA parlance, the identifiers are called NAME, NAME2, and ID, and are presented in that order in the file.

Following the identifiers is a number representing the type and length of the object. In all cases, the number itself represents the number of points that make up the object. Regions have at least four points, lines at least two, and points may only have one. To identify an object as a region versus a line, regions use a positive number, while lines are identified by a negative number.

In a BNA file, latitude/longitude coordinates must be represented in decimal degrees, with up to six decimals of precision. Longitude is the x coordinate, and latitude is the y coordinate. Western longitudes and southern latitudes are always negated. This means that for the U.S., longitudes should be negated, while latitudes should be positive. User coordinates should be entered as whole numbers, as the decimal portion is ignored.

The PIP Transform supports BNA files that are delimited with commas. All identifiers should be quoted. Each feature in a BNA file is made up of several lines of information. For use in the PIP Transform, the lines should be formatted as shown in the examples. This is the format used by ESRI's Atlas Import/Export product, which creates BNA files.

The basic format for objects in a BNA file is as follows:

```
"NAME", "NAME2", "ID", type/length
x1, y1
x2, y2
.
.
xn, yn
"Next object", " ", " ", type/length
x1, y1
.
.
xn, yn
```

Region Definitions

If the type/length is a positive number of four or greater, then that feature is a region and is followed by that number of coordinate pairs. In a BNA file, regions are always “closed”. That means that the last point in the region is the same as the first. Therefore, a triangle is expressed with four coordinates, the first and last coordinates being the same point.

It is possible for a region to contain Island or Lake polygons. In a BNA file, islands and lakes have an identical representation. The basic representation is as follows:

```

"Object with Island or Lake", "NAME2", "ID", n
x1, y1
.
.
x1, y1
xx1, yy1
.
.
xx1, yy1
x1, y1

```

The second xx1, yy1 pair closes the island or lake, and the final x1, y1 closes the main polygon. As you look at this structure, you can see that a “connecting line” is defined between the main polygon and the island or lake. Products that support BNA files, such as the PIP Transform, can detect this line during import, and can effectively remove it when stored.

For more information on islands or lakes in a BNA file, contact ESRI.

Line Definitions

If the type/length is negative, and the absolute number is two or greater, then that feature is a line and is followed by the absolute number of coordinate pairs. The basic structure for a line object is similar to a region, but the line does not have to have the last coordinate equal to the first. The basic representation for a line is as follows:

```
"NAME", "NAME2", "ID", -n  
x1, y1  
x2, y2  
.  
.  
xn, yn
```

In a BNA file, lines do not have islands or lakes.

Point Definitions

If the type/length is 1, then that feature is a point and is followed by a single coordinate pair. The basic representation for a point is as follows:

```
"NAME", "NAME2", "ID", 1  
x1, y1
```

Appendix C

MapInfo MIF/MID Format

This appendix summarizes the subset of MapInfo's Data Interchange Format known as the MIF/MID format supported by the Point-in-Polygon Transform.

In this appendix:

- Introduction
- MIF Files
- Region Definition
- Polyline Definition
- Point Definition
- MID Files

Introduction

The Point-in-Polygon Transform supports a subset of MapInfo's Data Interchange Format, also referred to as MIF/MID format. This section summarizes the format as supported by the PIP Transform. For a more complete description of the format, contact MapInfo.

The MIF/MID format is comprised of two files, one with an extension of MIF, the other with the same name, but the extension MID. The MIF file contains the actual points that represent the objects, while the MID file contains the text information about the objects.

MIF Files

MIF files contain a tokenized header, followed by a DATA section. The PIP Transform uses version 1 or version 2 files, and requires the comma delimiter, which is the default for MapInfo. If a DELIMITER token is present in the file, the file is not imported.

The next part of the file header contains the definitions of the columns in the MIF file. Feature names, as well as numeric data about the features, are stored here. The PIP Transform uses one of these columns as the identifier to store with the object.

The column definitions begin with the COLUMN token, followed by the number of columns in the MIF file. Following the COLUMN token is a single line entry for each column. The first entry on the line is the column name, the second is the column type. The example below indicates three columns in the MIF file:

```
COLUMN 3
FEAT_NAME    char (15)
POPULATION   integer
AREA         decimal (8,4)
```

The PIP Transform can use any column type as the name, but char is the most common, since that represents a string. The number in parentheses following char indicates the largest string; the string for the PIP Transform cannot be more than 127 characters. For a complete listing of column types, contact MapInfo.

The token `DATA` indicates that the header is complete, and the listing of objects follows. For each object, a number of attributes may be listed, such as `PEN` and `BRUSH`. The PIP Transform ignores these settings. MapInfo supports a variety of object types, though only a few are used to represent geographical objects. The PIP Transform supports three object types – Region, Polyline, and Point.

In a MIF file, latitude/longitude coordinates must be represented in decimal degrees, with up to six decimals of precision. Longitude is the x coordinate, and latitude is the y coordinate. Western longitudes and southern latitudes are always negated. This means that for the U.S., longitudes are negated, while latitudes are positive. User coordinates are entered as whole numbers, as the decimal portion is ignored.

Region Definition

A region consists of one or more polygons. Each polygon is uniquely defined. In a MIF/MID file, you do not need to “close” a polygon. The last point in the polygon automatically attaches to the first point in a polygon with a line segment. The format for a region object is as follows:

```
REGION #pgons
numpts1
x1 y1
x2 y2
.
.
numpts2
x1 y1
x2 y2
.
.
```

Polyline Definition

A polyline (`PLINE`) is a line that contains any number of points to define its shape. The number of points in the line follows the `PLINE` identifier. Below is the format for a `PLINE` object.

```

PLINE numpts
x1 y1
x2 y2
.
.

```

Point Definition

A point is defined as a single coordinate pair. Below is the format for a point object.

```
POINT x y
```

MID Files

MID files list the data, such as name, for objects defined in the MIF file. There is one entry in the MID file for each object in the MIF file. The entry in the MID file must be in the same order as the objects appear in the MIF file. The data in the MID file is delimited by commas, unless otherwise configured by the `DELIMITER` statement in the MIF file. The PIP Transform supports only the comma delimiter.

As an example, remember the column definitions used earlier in the MIF file:

```

COLUMN 3
FEAT_NAME char (15)
POPULATION integer
AREA decimal (8,4)

```

With this column definition, and assuming comma delimiters, sample entries in the MID file appear as:

```

First Feature,180232,1000.2345
California,4345105,56023.3569
Last Feature,12643,136.4000

```

Note: Columns are separated by a single comma. There is no header or other information contained in the MIF file.

For a complete description of MIF/MID file formats, contact MapInfo.

Glossary

Spatial Analysis Terms

B

Buffer

An area drawn around an object. Buffers test whether points are "near" other objects, such as points, lines, or polygons.

C

Closest Site

Determines distances between two sets of points.

G

Geocoding

The process of assigning data based upon location information. A common example of geocoding is the assignment of latitude and longitude values to an address location.

GIS

Geographic Information System. A computer-based tool for enhancing geographic data by analyzing both the physical location in space and the set of characteristics associated with a location.

P

Point-in-Polygon Analysis

An analysis that determines within which polygon or polygons a point lies.

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